LOT SYSTEMS CORPORATION
DOCUMENT: LOT-COSMO-COMPUTER-v1
TITLE: COSMO® Computer — A Physical Companion for the LOT® Operator
CLASS: RESTRICTED // S-2 EYES
S-2: VADIK MARMELADOV
DATE: 2026-08-05
VERSION: 1.0 — DEVELOPMENT START (PLAN + BOM + ROADMAP, PRE-MANUFACTURE)
STATUS: PLANNING — no PCB spun, no PCBWay order placed

00 // READING LOG — SOURCES THIS DOCUMENT IS BUILT ON

This is not the first hardware document at LOT. It is the second physical product spec, and it exists specifically to NOT collide with the first. Before writing a line of this plan, the following were read in full:

docs/benchmark/LOT-MANIFEST.md
Section 01 records a prior entry — "COSMO Hardware | brave-lamport-t9z5u8 | BEST | +2610 lines | COSMO® Cube — complete hardware computer design v1.0." That branch no longer exists on the remote (Section 04 note, 2026-06-27: "incorporated into master in prior sessions"), and no surviving file under that description was found on master. The name is real, the content is lost. This document is the reconstruction — written fresh, from S-2's 2026-08-05 brief, not a recovery of the old branch's bytes.

docs/corporate/LOT-CUBIQ-QUANTUM-CUBE-v0.md
The sibling hardware track. CUBIQ™ is a 45mm nano-ceramic cube that HOPS — a notification body with no screen, no camera, motion as its only language. Its own Section 00 already drew the line this document must hold: "CUBIQ™ is LOT®'s object: a notification body, not a computer... The two are related by lineage (father/son, LOT®/COSMO®) and should share no naming collision going forward." This document is the COSMO® side of that lineage — the object CUBIQ's authors were naming in advance.

docs/corporate/LOT_ROBOTICS_COSMO.md
Establishes the COSMO® brand itself: "COSMO® is the robotics product line of LOT Systems, named after Kuzya Cosmo Marmeladov." Names the Benchmark Arbitrage® gate, the Soul Sync Protocol™, and the Phase 3 revenue line ("COSMO® Hardware | 2028-2029 | $2,500-$5,000 per unit"). This document does not repeat that ethical/IP framing — it assumes it and builds the first physical SKU underneath it.

docs/corporate/CQGS-WHITE-PAPER-SNAPSHOT.md
Section III, Layer 4 — Memory Arc: "12mo+ hardware" is the named milestone at which a subscriber's engagement graduates into a physical object. The Products row already names "Quantum Cube | Bioelectric hardware, haptic feedback, nano-ceramic, piezoelectric" — that line became CUBIQ. No line in CQGS yet named a device with a screen, a camera, and a button. This document opens that line.

docs/corporate/LOT-CUBIQ-OPERATOR.md
Section 04, "AI-Driven Physical Product Delivery," and Section 07 Phase 4 ("Physical Extension, Days 90+") both describe the AI choosing WHAT/WHEN/HOW to reach the operator physically. CUBIQ answers that with motion. This document answers it with a message the operator can read.

S-2 brief, 2026-08-05 (verbatim, 19-point logic list + free text)
The controlling input for this document. Every section below traces back to a numbered point in that brief; Section 09 is the point-by-point traceability table.

No prior surviving document specifies a screen-bearing, camera-bearing, LOT-API-connected device, a PCBWay manufacturing plan, or a components buying list. This document is that specification, v1.0.

01 // WHAT THE COSMO® COMPUTER IS AND IS NOT

NAME (WORKING) COSMO® Computer — "the Puck." Formal SKU TBD at v1.0 close (Section 08 opens the naming decision; this document does not lock a final consumer name).

IT IS
- A flat, silver, two-part stainless-steel object that sits on a desk, wirelessly charges, and shows the operator ONE short line of text when the LOT® Quantum Intent Engine (QI·46) decides they need to see it —

"Coffee time!" being the brief's own example.

- A small always-on edge computer, not a peripheral. It runs its own firmware, holds its own LOT API session, and can operate for a bounded window fully offline (Section 05) — it does not require the operator's phone or laptop to be open to receive a notification.
- A physical extension of the Log tab already live at lot-systems.com — the single hardware BUTTON (Section 04) writes an event back into the same Logs.tsx military-handler pipeline that already renders journal_entry, badge_unlock, and cohort_signal events (verified in src/client/components/SystemProgressWidget.tsx and src/server/routes/api.ts /logs endpoint family). This is the one place in this document where "connected to the LOT site" is not aspirational — the target pipe already exists in the running application.
- A camera-and-sensor node that feeds the same Calibration Loop CQGS already describes (Layer 1) — weather, presence, and an operator-initiated photo capture become deliberate + passive inputs, exactly the two input classes the Loop already expects.

IT IS NOT

- CUBIQ™. CUBIQ moves; COSMO® Computer does not actuate. No motor, no actuator, no jump. If a future unit needs to move, that capability belongs to the CUBIQ roadmap (v1-v3, already scoped), not this one.
- A general-purpose consumer computer. It runs one application: the LOT® client. It is not a development board, not a tablet, not a phone replacement. Sideloading is out of scope for v1.
- A phase-1 mass-market SKU. Per LOT_ROBOTICS_COSMO.md's own revenue table, COSMO® Hardware is a 2028-2029 line. This document plans the 100-unit v1.0 pilot run (Section 06) that makes that later phase possible — not a consumer launch.

THE PRINCIPLE

Ship the smallest true object first. A silver square that charges wirelessly, shows one line of AI-chosen text, takes one photo on demand, and writes one button-press back to the Log tab is a complete v1.0. A device that also tries to be a general computer, a security camera, and a weather station with full historical charting before the single notification loop is proven end-to-end is not a v1.0 — it is a fundraising deck.

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02 // PHYSICAL FORM (BRIEF POINTS 3, 4, 17, 18)

FOOTPRINT — 40mm x 40mm x 5mm — "a flat silver square 4x4cm x 5mm height" (brief point 4), read literally as the closed, assembled dimension. 5mm is tight for the stack in Section 03; Section 07 (roadmap) records where the margin comes from.

BODY — Two-part stainless-steel shell (brief point 3):

- TOP SHELL — polished stainless steel (brief point 17). Mirror-finish face, no printing, no visible fasteners. This is the face that sits toward the room — an object, not a gadget.
- BASE SHELL — the working face (brief point 18): camera, screen, and button live here, bead-blasted matte stainless around the screen window to cut glare. The base also carries the wireless-charge coil window (Section 03) and rests against the charging pad.

Two-part construction is a manufacturing decision, not a cosmetic one — it is the only way to seal a wireless-charge-compatible metal(*) enclosure, route a camera and display through 5mm of stack, and still hand-assemble 100 units without a laser-welder in the pilot run. See Section 06 note on stainless-steel Qi charging.

DISPLAY — Screen, base face (brief point 18, and the brief's own close: "Simple screen to show autonomous notifications ... e.g., 'Coffee time!'"). v1.0 target: monochrome or 3-color e-paper, ~1.5-2", inside the polished-stainless window on the base shell. E-paper is chosen over an LCD/OLED for three reasons that all trace to existing LOT doctrine, not just battery life:

1. It holds an image with zero draw current between refreshes — matching the CUBIQ anti-feed thesis

(LOT-CUBIQ-QUANTUM-CUBE-v0.md Section 04): "a blinking light is a screen substitute... this preserves the anti-feed thesis." A screen that is usually blank and occasionally states one fact, once, is the OPPOSITE of a live feed.

2. It fits the 5mm stack — e-paper modules run thinner than backlit LCD at this size class.
3. It reads in direct light on a desk near a window, where a phone screen washes out.

CAMERA — Base face, opposite the screen window (brief point 5, 18). Fixed-focus module, no gimbal, no zoom — one forward-facing lens for operator-initiated capture only (Section 05). No always-on recording; the camera wakes on the BUTTON action or an explicit LOT API command, never on a timer or a motion trigger. This is a deliberate restriction, not a v1.0 hardware limitation to be lifted later — see Section 08 gate.

BUTTON — Base face, single mechanical button, labeled "Copy" (brief point 16). Section 04 specifies its full signal path.

FINISH NOTE (*) — True 316-grade stainless is not RF-transparent — a solid stainless base shell will attenuate the Qi inductive charge field and any wireless radio inside it (Section 03 BOM). v1.0 resolves this with a machined non-metal (ABS/PC composite, stainless-finish PVD coating) insert ring at the coil and antenna window on the base shell, so the object reads as "stainless steel body" (brief point 3) to the hand and the eye while the charge and radio path stays physically unobstructed. This compromise is recorded here in the open — it is the single physical-form constraint most likely to move in v1.1 (Section 07).

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03 // ELECTRONICS ARCHITECTURE

COMPUTE — Espressif ESP32-S3 (dual-core, Wi-Fi + BLE, hardware JPEG/camera interface, 8-16MB flash + 8MB PSRAM variant). Chosen over a heavier SBC (Raspberry Pi Zero class) because the whole job — hold a Wi-Fi session, poll/receive one LOT API channel, drive one e-paper panel, decode one still image, debounce one button — fits a microcontroller-class part inside a 5mm-thick, wirelessly-charged, battery-buffered enclosure. This is the central engineering bet of v1.0; Section 09 traceability table logs it against brief point 6.

CAMERA — OV2640 or OV5640 module (2MP/5MP), SPI/DVP interface direct to the ESP32-S3's camera peripheral — the standard, off-the-shelf pairing (brief point 15: "AI grade off-the-shelf sensors" — read as "commodity-qualified parts, not custom silicon").

DISPLAY — GDEY0154D67 class 1.54" tri-color or monochrome e-paper panel, SPI. Driven by the ESP32-S3 directly; no separate display MCU.

WEATHER SENSOR — BME280 (temperature / humidity / barometric pressure), I2C — the same three-axis reading LOT-CUBIQ-OPERATOR.md Section 01 already lists under "PRESENCE: Weather and location context per session," now sourced locally on-device instead of purely from a remote weather API call. Satisfies brief point 14.

CHARGING — Qi-class wireless receiver coil + PMIC (e.g. BQ51013B-class receiver into a single-cell LiPo charge controller), matched to the base-shell insert ring (Section 02). Satisfies brief points 12 and 19 as ONE system, not two — "Charger" and "Wireless charger" in the brief are the same physical subsystem (receiver in the Puck, transmitter pad sold as the companion charging base, itself the flat surface the Puck rests on — directly mirroring the CUBIQ "the charging pad IS the table" pattern already on record).

BATTERY — Single-cell LiPo pouch, ~250-350mAh, the largest cell that clears the 5mm z-height stack alongside the coil and PCB. Buffers the unit through charge-pad gaps; not sized for all-day cordless operation — this is a desk object, expected to sit on its charging pad most of the time.

BUTTON — Single mechanical tactile switch under the "Copy" cap (Section 04), wired to a dedicated GPIO with hardware debounce.

CONNECTIVITY Wi-Fi (2.4GHz) for the LOT API connector (Section 05); BLE reserved for v1.0 provisioning only (pairing the unit to an operator's account via the existing lot-systems.com session, not a runtime data path).

PCB — Single 4-layer rigid board, sized to the 40x40mm footprint minus shell wall thickness — realistically ~34x34mm usable — manufactured via PCBWay (brief point 1; full spec in Section 06).

04 // THE BUTTON — "COPY" SIGNAL PATH (BRIEF POINT 16)

The brief specifies the button's label and its destination precisely: "Button as 'Copy' with a signal back to the site's Log tab on lot-systems.com." This is the one requirement in the brief that maps directly onto a pipeline already running in the codebase — not a new concept, a new event source into an existing one.

WHAT "COPY" MEANS

A single press captures the operator's current on-screen notification text (whatever is currently rendered on the e-paper, e.g. "Coffee time!") and pushes it — copies it — into the operator's LOT session as a logged acknowledgment. It is the physical equivalent of clicking a toast notification to dismiss-and-record it, except the object is not a screen the operator was already looking at.

SIGNAL PATH

```
BUTTON (GPIO interrupt)
    |
    v
FIRMWARE debounces, reads currently-displayed notification ID
    |
    v
LOT API CONNECTOR (Section 05) — POST device event, authenticated
to the operator's session
    |
    v
SERVER — new event type on the existing log-event pipeline
(src/server/routes/api.ts /logs family; event shape follows the
same pattern already handled by src/client/components/SystemProgressWidget.tsx's
military-handler dispatch — journal_entry, badge_unlock, cohort_signal,
etc. all resolve to one dispatch table keyed on event name)
    |
    v
LOGS.TSX — new military handler, working name COPY:
(device_notification_copy) — renders alongside the existing REC:,
BADGE:, COHORT:, VITALS: handlers already listed in
SystemProgressWidget.tsx's own build history
    |
    v
Operator's Log tab at lot-systems.com shows: COPY: "Coffee time!"
— device <id> — <timestamp>
```

WHY THIS IS THE RIGHT SCOPE FOR v1.0

One button, one event type, one new handler in an already-existing dispatch table. This is deliberately the smallest possible wiring of "hardware button -> site Log tab" — no new database table, no new subsystem, one row in a dispatch table that already has a dozen entries of exactly this shape. It is real engineering scope (Section 09 marks it ENGINEERING, not just CORPORATE), sized to be shippable inside a normal Benchmark cycle once firmware exists to call it.

05 // THE LOT API CONNECTOR (BRIEF POINTS 2, 6)

Brief point 6 says "Use LOT API connector"; brief point 2 says "Send a pager-like notification from an AI-powered site." These are the same subsystem read from two ends — the site pushes, the device receives, and the same channel carries the button's event back up (Section 04). v1.0 defines the connector as a single bidirectional channel, not two:

DOWN-CHANNEL (site -> device) — THE PAGER

QI·46 decides a notification should reach the operator (the same decision point LOT-CUBIQ-QUANTUM-CUBE-v0.md Section 05 already diagrams for CUBIQ: "QI·46 CALIBRATION LOOP -> Index of Systems (signal fires)"). Instead of driving an actuator, the signal is rendered to a short text string (<= ~20 characters at v1.0 e-paper size) and delivered to the device's provisioned session. Delivery is intentionally PAGER-CLASS — one short line, no rich payload, no images, no threading — because the brief's own language ("pager-like") is doing real design work here: a pager cannot become a feed. That constraint is a feature, not a v1.0 limitation.

UP-CHANNEL (device -> site) — TELEMETRY + THE COPY EVENT

Weather-sensor readings (Section 03), the operator-initiated camera capture, and the Section 04 button event all travel the same up-channel into the existing log-event pipeline. This mirrors the CQGS Calibration Loop's own two-class input model (deliberate + passive) — the button press and the camera trigger are DELIBERATE inputs; the weather reading is PASSIVE, ambient, continuous-ish (polled, not streamed, to protect battery).

TRANSPORT (v1.0)

HTTPS long-poll or short-interval poll over Wi-Fi, authenticated with a per-device token issued at BLE provisioning time (Section 03). A persistent WebSocket / push channel is the named v1.1 upgrade (Section 07) once the poll-based loop is proven reliable in the field — same "prove the primitive before extending it" discipline the CUBIQ document already applies to its own roadmap.

WHAT THIS DOCUMENT DOES NOT DO

It does not specify the exact REST/WS route names, payload schema, or auth token format — that is server-side engineering work for the companion document, Section 10 below, and belongs to a normal Benchmark ENGINEERING session against the live src/server/routes tree, not to a corporate plan document. What is fixed here is the SHAPE of the connector: one pager-class down-channel, one telemetry-class up-channel, both authenticated to one operator session.

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06 // MANUFACTURING PLAN — PCBWAY, 100-UNIT PILOT RUN (BRIEF POINTS 1, 13)

FABRICATOR

PCBWay (brief point 1) — chosen at the brief level, not re-litigated here. PCBWay's standard offering covers everything this run needs from one vendor:

- PCB fabrication (4-layer, Section 03 spec)
- SMT assembly (PCBA) — populate the ESP32-S3, camera module, PMIC, and passives
- CNC machining quote path for the two stainless-steel shell halves (Section 02) — PCBWay's CNC service handles small-run stainless machining, which keeps the whole pilot run (board + shells) inside one supplier relationship for v1.0, even though CNC and PCBA are ordered as separate line items in PCBWay's own ordering flow.

PCBWay instant-quote entry point:
https://www.pcbway.com/orderonline.aspx

PCBWay CNC machining entry point:
https://www.pcbway.com/rapid_prototyping.html

RUN SIZE

100 units (brief point 13) — sized as a pilot: enough to (a) qualify the SMT line and CNC tolerance stack at real volume, (b) put units in front of Purple/Black-tier operators per the Benchmark Arbitrage® gate already defined in LOT_ROBOTICS_COSMO.md, and (c) generate the first real BOM cost data before any Phase-3 ($2,500-$5,000/unit, per LOT_ROBOTICS_COSMO.md) pricing commitment is made. 100 is below PCBWay's usual small-batch PCBA break-even efficiency curve — expect the per-unit assembly cost to drop materially at the next run (500-1,000 units), which is exactly the kind of data point v1.0's pilot exists to produce.

ASSEMBLY SEQUENCE

1. PCBWay fabricates + assembles (PCBA) the 34x34mm 4-layer board, 100 units.
2. PCBWay CNC-machines 100 top shells (polished) + 100 base shells (matte + insert ring, Section 02) in 316 stainless steel.
3. E-paper module, camera module, battery, and Qi receiver coil hand-fit into the base shell at LOT

(not PCBWay) for the pilot run — 100 units is within hand-assembly range and keeps the delicate fit-up (Section 02's 5mm stack) under direct LOT QA rather than a remote line's tolerance.

4. Final shell-to-shell seal (screws or press-fit, decided at first physical prototype, not in this document) and firmware flash (Section 10) at LOT before shipment.

QA GATE — Every unit must pass, before shipment: (a) Qi charge acceptance on the reference charging pad, (b) one successful pager-class notification round-trip against a LOT staging account, (c) one Copy-button event observed landing in that staging account's Log tab, (d) e-paper refresh with no visible ghosting. This is the hardware analogue of the software Benchmark green gate (docs/benchmark/ protocol) — no unit ships red.

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07 // ROADMAP — v1.0 -> v1.1 -> v1.2 -> v2.0

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v1.0 — THE PUCK (THIS DOCUMENT)
Single ESP32-S3, e-paper pager display, fixed-focus camera, BME280 weather sensor, one Copy button wired to the Log tab, Qi charging through a non-metal insert ring, PCBWay 100-unit pilot run.
GATE: 100/100 units pass the Section 06 QA gate; 30 consecutive days of pager-notification delivery to at least 10 real Purple+ operator desks with zero unrecoverable firmware lockups.

v1.1 — PERSISTENT CHANNEL + TRUE STAINLESS RF PATH
Replaces the poll-based up/down channel (Section 05) with a persistent push connection. Opens material research into a genuinely RF-transparent stainless alloy or a thinner true-metal shell that doesn't need the Section 02 insert-ring compromise.
GATE: notification latency under 2s at the 95th percentile across a 50-unit fleet; insert-ring elimination prototyped in at least one material candidate.

v1.2 — MULTI-GESTURE PAGER VOCABULARY
CUBIQ ships four gestures from one hop primitive (NUDGE/HOP/LEAP/SETTLE). COSMO® Computer v1.2 does the display-side equivalent: beyond a single line of text, a small fixed icon set (weather glyph, badge-unlock glyph, memory-question glyph) so the pager channel can communicate signal CLASS at a glance, not just content — without crossing into the "screen substitute" territory the anti-feed thesis warns against (Section 02).
GATE: icon set held to <= 6 total glyphs; user-tested for correct recognition at 90%+ without a legend.

v2.0 — COSMO® / CUBIQ CONVERGENCE (RESEARCH TRACK, NOT SCHEDULED)
Named here so v1.0-v1.2 electronics and enclosure choices are made with it in mind, not foreclosed: a future object that carries the Puck's screen/camera/API stack (this document) inside a body that can also perform CUBIQ's controlled hop (LOT-CUBIQ-QUANTUM-CUBE-v0.md Section 03). No mass, power, or mechanical-compatibility claim is made here — this is a horizon note, exactly as CUBIQ's own Section 06 treats levitation as a v.3 horizon rather than a build milestone.

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08 // NAMING, GATE, AND OPEN DECISIONS AT v1.0 CLOSE

NAMING OPEN — "COSMO® Computer" is a working name throughout this document. It correctly carries the COSMO® mark (LOT_ROBOTICS_COSMO.md) and correctly avoids the CUBIQ™ collision (Section 00). A final consumer-facing name is not locked in this document — record candidates in a future session, do not invent one here to avoid a false sense of finality.

CAMERA GATE — No always-on capture, no motion trigger, no remote-initiated capture without an explicit operator action in v1.0. Loosening this gate is an explicit, separate decision for a future document — it is recorded here as a boundary specifically so no future session drifts past it silently.

STAINLESS/RF

COMPROMISE

Section 02's insert-ring approach is the current answer, flagged as the most likely thing to change in v1.1 (Section 07). Not treated as settled.

BOM AND ROADMAP ANALYSIS

Full component-by-component buying list, unit-cost roll-up for the 100-unit run, and supplier links live in the companion document (Section 10, item 2) — kept separate per brief point 11 ("Separate documents") rather than inlined here.

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09 // TRACEABILITY — BRIEF POINT TO DOCUMENT SECTION

#	BRIEF POINT	SECTION
1	PCB Way	06
2	Pager-like notification from AI-powered site	05 (down-channel)
3	2 parts stainless steel body	02
4	Flat silver square 4x4cm x 5mm height	02
5	Camera	02, 03
6	Use LOT API connector	05
7	Result in PDF manuals	10 (item 3)
8	Compress the information in each session	this doc's own Section 00/09 traceability discipline; see also LOT-SR report, block 06
9	Firmware documents	10 (item 1)
10	Software to connect with firmware	10 (item 1, LOT API connector client half)
11	Separate documents	10 (doc split)
12	Charger	03
13	100 units run	06
14	Weather sensor	03
15	AI grade off-the-shelf sensors	03
16	"Copy" button -> Log tab signal	04
17	One side polished stainless steel	02
18	Other side: camera, screen, button	02, 03, 04
19	Wireless charger	03

Free-text close (screen shows notifications, e.g. "Coffee time!")   02, 05

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10 // COMPANION DOCUMENTS (BRIEF POINT 11 — SEPARATE DOCUMENTS)

Per the brief's own instruction to keep documents separate, this plan does not inline firmware, software, or the BOM. Companion set, v1.0:

1. docs/technical/LOT-COSMO-COMPUTER-FIRMWARE-v1.md
   On-device firmware architecture: boot, provisioning, e-paper driver, camera capture path, button debounce + Copy event, weather-sensor poll loop, LOT API connector client (device side).

2. docs/corporate/LOT-COSMO-COMPUTER-BOM-v1.md
   Full components buying list — part numbers, quantities for the 100-unit run, supplier links, and a unit-cost roll-up.

3. docs/technical/LOT-COSMO-COMPUTER-SOFTWARE-v1.md
   Server + client (lot-systems.com) side of the LOT API connector: the down-channel notification dispatcher, the up-channel log-event ingestion (Section 04's Copy path), and the Log tab handler addition.

4. PDF manuals (brief point 7) — generated from documents 1-3 and this plan, filed alongside their source .md in the same folders.

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END LOT-COSMO-COMPUTER-v1

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